

Adam Bouafia

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Summary

Cloud and DevOps engineer with an MSc in Computer Science (double degree, VU Amsterdam and Università dell'Aquila). Hands-on experience with Azure, Kubernetes, and CI/CD pipelines, including production Azure infrastructure built entirely with Infrastructure as Code. Working through the Linux Foundation Advanced Cloud Engineer program and preparing for the CKA. Focused on automation, observability and green computing.

Experience

Master's Thesis Intern, Vrije Universiteit Amsterdam – Amsterdam, Netherlands Mar 2025 – Jan 2026

- Thesis: LogPress
- Semantic Log Compression System
- Built a semantic log compression system processing 2.4M+ logs with a 22.33x compression ratio and 100% lossless reconstruction
- Implemented a Lazy Reconstruction query system answering in 21-111ms without full decompression
- Containerised the pipeline with Docker and automated CI/CD via GitHub Actions

DevOps Engineer, Vivieb – Pescara, Italy Nov 2023 – Feb 2024

- Optimized Kubernetes cluster energy efficiency using Kubegreen
- Built monitoring dashboards with Grafana and InfluxDB
- Implemented automated deployment pipelines

Software Developer, Euromed Organization – Paris, France July 2022 – Dec 2022

- Developed Dhayen mobile security app for national awareness campaign
- Implemented CI/CD pipeline for mobile app deployment

Professional Development

The Linux Foundation - Advanced Cloud Engineer IT Professional Program Mar 2026 – present

- Eight-course cloud native track: Containers (LFS253), Kubernetes (LFS258), Prometheus (LFS241), Fluentd and Fluent Bit (LFS242), Service Mesh (LFS243), Helm (LFS244)
- Built and operated a three-node kubeadm cluster on libvirt/KVM: cluster build and upgrade, etcd backup and restore, control plane recovery, RBAC, CNI networking and live failure troubleshooting
- Preparing for the CKA (Certified Kubernetes Administrator) exam

Microsoft Azure Certification Path In progress

- Preparing for AZ-104 (Azure Administrator Associate), followed by AZ-400 (Designing and Implementing Microsoft DevOps Solutions) toward the DevOps Engineer Expert credential

Education

Vrije Universiteit Amsterdam / Università dell'Aquila, MSc in Computer Science (Double Degree)

- Specialization: Software Engineering & Green IT

Projects

FirstCall - AI First Responder for Kubernetes Incidents

- Built for the AI Accel hackathon; ships as a Helm chart on any conformant cluster
- Collects first-15-minutes incident evidence (pod spec, events, logs, metrics, rollout diff) and redacts secrets before anything leaves the cluster
- Diagnoses with open-weight models over an OpenAI-compatible API, backed by quoted evidence
- Read-only by default: fixes are allow-listed, server dry-run and applied only after human approval
- Prometheus metrics, Grafana dashboard, and CI running every change against a disposable kind cluster

repo-sentinel - Autonomous Repository Auditor

- Agent on the Claude Agent SDK auditing repos for failing CI, outdated dependencies and upstream drift
- Opens guarded fix PRs on sentinel/* branches; PR rules enforced in code, not in the prompt
- No shell and no local file access: read-only tools scoped to the repo under audit
- Keeps state between runs so findings are marked new, open since a date, or resolved

GitHub Contributor: Kubernetes-Certified-Administrator (walidshaari, 4.2k stars)

- Updated the CKA study guide to the 2026 curriculum (Kubernetes v1.35)
- Added a troubleshooting guide, exam-day tactics and study method docs
- Built a kubeadm Vagrant lab for hands-on cluster practice
- Merged upstream as PR #63; added as repository contributor by the maintainer

Azure Landing Zone

- Production Azure infrastructure with hub-spoke topology using 100% Bicep IaC
- Azure Firewall, Bastion, NSG, Key Vault with RBAC
- Multi-stage CI/CD pipeline (Azure DevOps) with validation, what-if, and approval gates
- Automated cost optimization scripts saving 70% on dev infrastructure

Greenmining

- Python library published on PyPI for Green IT research
- 124 sustainability patterns from Green Software Foundation
- Multi-backend energy measurement (RAPL, CodeCarbon)
- GitHub Actions CI/CD with automated testing and deployment

LCT - LLM Comparative Tool

- Open-source framework evaluating 500K+ HuggingFace models
- Dockerized deployment with CI/CD pipeline
- Real-time energy profiling and carbon footprint analytics

CO2 Consumption Monitor

- GNOME Shell extension published on GNOME Store
- Real-time CO2 emissions tracking with ElectricityMaps API integration
- Per-application monitoring and historical data export

IoT Monitoring System

- Real-time monitoring using MQTT, Node-RED, InfluxDB, Grafana
- Automated alerts via Telegram integration
- Linux-based deployment on Raspberry Pi

Skills

Cloud Platforms: Azure

Containerization & Orchestration: Kubernetes (kubeadm, RBAC, etcd operations), Docker, Helm, Knative

Infrastructure as Code: Terraform, Bicep, Ansible

CI/CD & GitOps: Azure DevOps, GitHub Actions, GitLab CI, Jenkins, ArgoCD

Operating Systems: Linux (Fedora, Ubuntu, RHEL), Windows

Scripting & Programming: Python, Bash, Java, JavaScript, SQL

Monitoring & Logging: Prometheus, Grafana, InfluxDB, ELK Stack, Fluentd, Fluent Bit

Version Control: Git, GitHub, GitLab

Languages

English: Native

Dutch: Beginner

Arabic: Native

French: Proficient

Italian: Beginner